

```

101   status?: 'Pending' | 'Accepted' | 'Rejected' | 'Completed';
102   createdAt?: Timestamp | Date;
103 }

```

Listing 8.6: Type Definitions (types.ts)

The type definitions ensure that all data flowing through the application conforms to a strict schema. The `LocationData` interface standardizes geolocation data with optional address fields. Enums like `ListingStatus` and `WasteReportStatus` enumerate all valid states, preventing invalid status assignments at compile time.

8.7 Location Service

The location service module provides GPS-based geolocation functionality and reverse geocoding:

```

1  export interface Location {
2    latitude: number;
3    longitude: number;
4  }
5
6  export function getCurrentLocation(): Promise<Location> {
7    return new Promise((resolve, reject) => {
8      if (!navigator.geolocation) {
9        reject(new Error('Geolocation is not supported'));
10       return;
11     }
12     navigator.geolocation.getCurrentPosition(
13       (position) => {
14         resolve({
15           latitude: position.coords.latitude,
16           longitude: position.coords.longitude,
17         });
18       },
19       (error) => { reject(error); },
20       { enableHighAccuracy: true, timeout: 15000, maximumAge: 0 }
21     );
22   });
23 }
24
25 export async function reverseGeocode(
26   lat: number, lng: number
27 ): Promise<string> {
28   try {

```